KAPASA MAKASA UNIVERSITY

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

BACHELOR OF SCIENCE IN CYBER SECURITY

CYS 425 – Mobile Device and Security and Forensics

Date: 15th May, 2026 **Duration:** 1 hr 30mins

Marks: 60 **Time:** 11 hours

Instructions: There are two (2) sections **A** and **B**. Answer the compulsory question in **Section A**. Choose and answer any two (2) questions in **Section B.** Use a separate answer sheet to answer all questions.

SECTION A

[QUESTION 1]

You are a forensic investigator with the Zambia Police Service investigating a kidnapping case. The suspect's mobile device has been seized, but the suspect claims they were in a different city at the time of the crime. You need to use mobile network forensics to verify or disprove their alibi.

(a) The suspect's device was active during the time of the kidnapping.
   i. What are Call Data Records (CDRs) and what specific information do they contain that could help establish the suspect's location? (4 Marks)

(b) The crime occurred in an area with 4G coverage.
   i. How does LTE network analysis differ from GSM analysis in terms of the type of data available? (2 Marks)
   ii. Explain two advantages of LTE over GSM for forensic investigation. (2 Marks)

(c) The suspect's lawyer argues that cell tower logs are not accurate enough to pinpoint location.
   i. Describe how cell tower logs work and explain their limitation for precise location tracking. (3 Marks)

(d) The case also involves a 5G-enabled device that the suspect claims was at home during the crime.
   i. What advanced forensic capability does 5G offer that previous generations (GSM, LTE) lack, specifically regarding location tracking? (2 Marks)
   ii. Explain how this enhanced location data could help your investigation. (2 Marks)

(e) During your analysis, you discover that the suspect's device used WhatsApp for communication instead of standard SMS.
   i. What network logs would still be available, and what metadata could you analyze? (3 Marks)

(f) You need to present network evidence in court.
   i. List two (2) types of metadata you would extract from network logs to reconstruct the suspect's communication patterns. (2 Marks)

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SECTION B

[QUESTION 2]

You are a digital forensics expert examining a mobile device from a corporate espionage case. The suspect allegedly connected to the company's Wi-Fi network and transferred confidential files via Bluetooth to an accomplice in the parking lot.

(a) You begin by examining Wi-Fi logs on the suspect's device.
   i. What specific information would you look for in the Wi-Fi logs to prove the device connected to the company's network? (3 Marks)
   ii. How can Wi-Fi access point data be used for location tracking? (2 Marks)

(b) During the investigation, you capture unencrypted network traffic using a wireless sniffer.
   i. Name two tools that could be used for Wi-Fi sniffing and describe what data they can capture. (4 Marks)

(c) The suspect's Bluetooth logs show a pairing with an unknown device at the estimated time of the data transfer.
   i. What information is typically stored in Bluetooth logs? (3 Marks)
   ii. Describe two forensic challenges you face when analyzing Bluetooth evidence. (4 Marks)

(d) The defense claims that the Bluetooth connection evidence is unreliable because of its limited range.
   i. What is the typical range limitation of Bluetooth, and how does this actually strengthen your case? (3 Marks)
   ii. Name one (1) forensic tools that could be used to discover and analyze Bluetooth devices in an investigation. (1 Marks) BTscanner

[QUESTION 3]

You are a forensic investigator examining a suspect's phone in a drug trafficking case. The suspect has deleted all messages, call logs, and photos before the device was seized. You believe critical evidence can still be recovered.

(a) Define file carving and explain how it can be used to recover deleted images from the suspect's device. (4 Marks)

(b) The suspect's SMS messages were deleted. Explain three techniques you could use to recover deleted text messages. (3 Marks)

(c) You attempt logical recovery first but it fails.
   i. What is logical recovery and why might it fail in this situation? (3 Marks)
   ii. What is physical recovery, and how does it differ from logical recovery? (3 Marks)

(d) Explain the concept of unallocated space and why it is critical for recovering deleted data before it is overwritten. (4 Marks)

(e) Name three (3) forensic tools that can recover deleted data from mobile devices and briefly describe each. (3 Marks)

[QUESTION 4]

You are the incident response manager for a large corporation that has just suffered a mobile security breach. Several employees' mobile devices have been compromised by malware that appears to be exfiltrating corporate data.

(a) You activate the incident response plan. List and describe the six phases of incident response in the correct order. (6 Marks) PACERL

(b) During the detection phase, you identify the malware on three devices.

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i. What immediate containment steps should you take to prevent further data loss? (3 Marks)
ii. Why is it important to isolate compromised devices before beginning eradication? (2 Marks)

(c) The company does not have a formal mobile device security policy. You are asked to draft one.
i. List five elements that should be included in a mobile security policy. (5 Marks)

(d) Recommend four (4) best practices for securing mobile devices that you would include in employee training. (4 Marks)

[QUESTION 5]

You are presenting digital evidence from a mobile forensic investigation in a murder trial. The defense attorney is challenging the admissibility of your evidence, claiming it was mishandled.

(a) The judge asks you to explain the four (4) key requirements for digital evidence to be admissible in court. List and briefly describe each requirement. (8 Marks)

(b) The defense claims that cloud-stored evidence should be excluded because it exists in a different legal jurisdiction. Evaluate the validity of this argument. (4 Marks)

(c) During cross-examination, you are asked about your ethical obligations as a forensic expert.
i. Define the principle of confidentiality in forensic investigations. (2 Marks)
ii. What is the principle of objectivity, and why is it critical for expert witnesses? (3 Marks)

(d) Name three laws or legal frameworks that govern digital evidence handling. (3 Marks)

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